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Modelling The 200-Mile Fishing Limit With a Reaction-Diffusion System

Abubakar Othman (F221611)

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Department of Mathematical Sciences

Abstract

This report studies the effect of the 200-mile fishing limit on fish population using one dimensional reaction-diffusion model with spatially varying harvest. A single species benchmark and a two-species competition model are developed, with lower harvest inside the Exclusive Economic Zone (EEZ) and higher harvest outside it. The models are solved numerically using a finite-difference method with no-flux boundary conditions.

The simulations show that the 200-mile limit changes both the spatial distribution of biomass and the long run balance between protected and unprotected waters. In the single species case, lower harvesting inside the EEZ supports higher biomass in the long run, while diffusion allows biomass to leak into more heavily harvested offshore waters. In the competing species case, coexistence is possible, but the biomass now depends strongly on competition strength, harvest pressure and starting location. Sensitivity analysis shows that long term outcomes are most affected by the competition coefficients and harvest pressure, whilst diffusion has a weaker effect under the baseline parameter set. Varying the harvest outside the EEZ showed a trade-off between long-run biomass and cumulative catch, with over harvesting reducing population size and long term yield.

The results suggest that the 200-mile limit can protect biomass locally, but the long term effects depends critically on competition and harvest strength.

Key Words: Reaction-Diffusion Model, Exclusive Economic Zone (EEZ), 200-Mile, Fishing Limit, Competing Species, Harvesting, Diffusion, Sensitivity Analysis

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1 Introduction

1.1 Real-world Context

In 1976, the Parliament of the United Kingdom passed the Fishery Limits Act 1976 following the Cod War, a series of confrontations between Iceland and the United Kingdom over fishing rights in the North Atlantic around Iceland. The act implemented the extension of fishing waters under the European Community's Common (EEC) Fisheries Policy into British law. Contributors (2025)

The act extended the fishing limits from 12 to 200 nautical miles and was in force for the members of the EEC and nine other countries, excluding Iceland. The UK had also accepted Iceland's establishment of a 12-nautical-mile Icelandic exclusive zone around its shores and a 200-nautical-mile limit beyond which other nations' fleets required Iceland's permission. Contributors (2021)

The 200-nautical-mile limit is an Exclusive Economic Zone (EEZ). An EEZ is prescribed by the United Nations Convention on the Law of the Sea, under which a state has exclusive rights to the exploration and use of marine resources within its EEZ. Contributors (2019a)

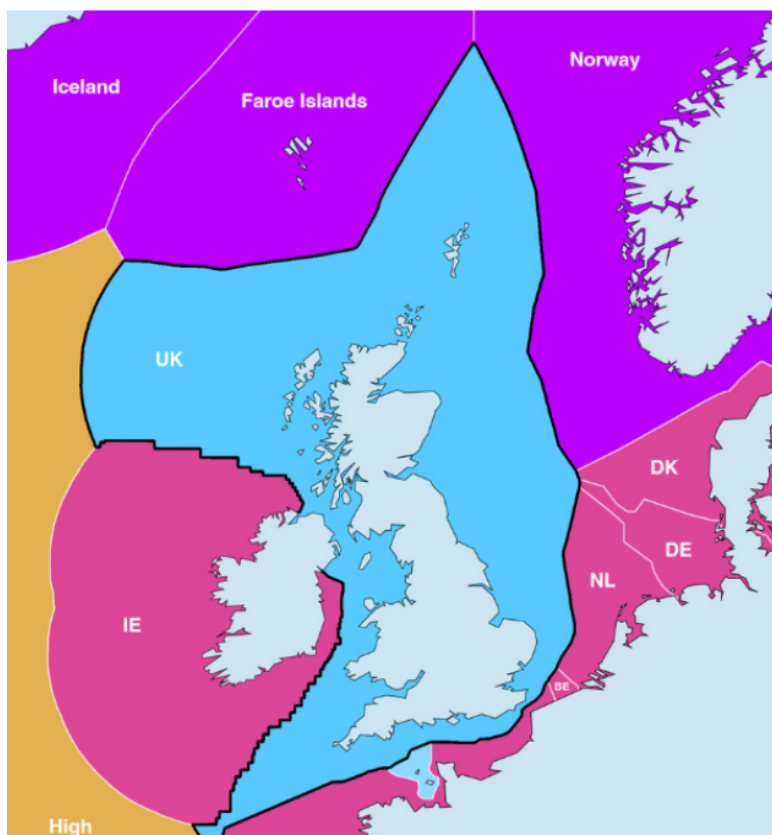


Figure 1: Map of the UK and the surrounding high seas, illustrating the policy distinction between waters under national jurisdiction and offshore waters beyond the 200-mile-limit.

Organisation (2025)

1.2 Why a spatial model is needed

We can leverage mathematical models to provide a systematic way to describe and analyse overall fish growth, the effects of harvesting on the population, and the distribution of fish.

A model that accounts for the spatial distribution of the population is necessary for analysing the system, as the 200-mile limit is a spatial boundary rather than a time-dependent ordinary differential equation that describes overall population growth, not migration. By introducing a one-dimensional model, we can capture how fish densities vary with time and space, and the effect of the boundary on the fish population through stricter harvesting within the 200-mile fishing limit.

Additionally, it is important that we consider a multi-species system. A single-species model provides a useful baseline for evaluating the effect, but it misses the important effects of competing species. We also need to consider how the harvest affects both the spatial and temporal distributions of fish in the system.

This report focuses on two species at sea and on applying fishing regulations to them. We carry out this analysis using the Lotka-Volterra model and the fish population distribution model.

2 Background

2.1 Logistic Growth

The logistic model is a standard model for population growth under limited resources.

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right) \quad (1)$$

where N is population size, r is the intrinsic growth rate, and K is the carrying capacity.

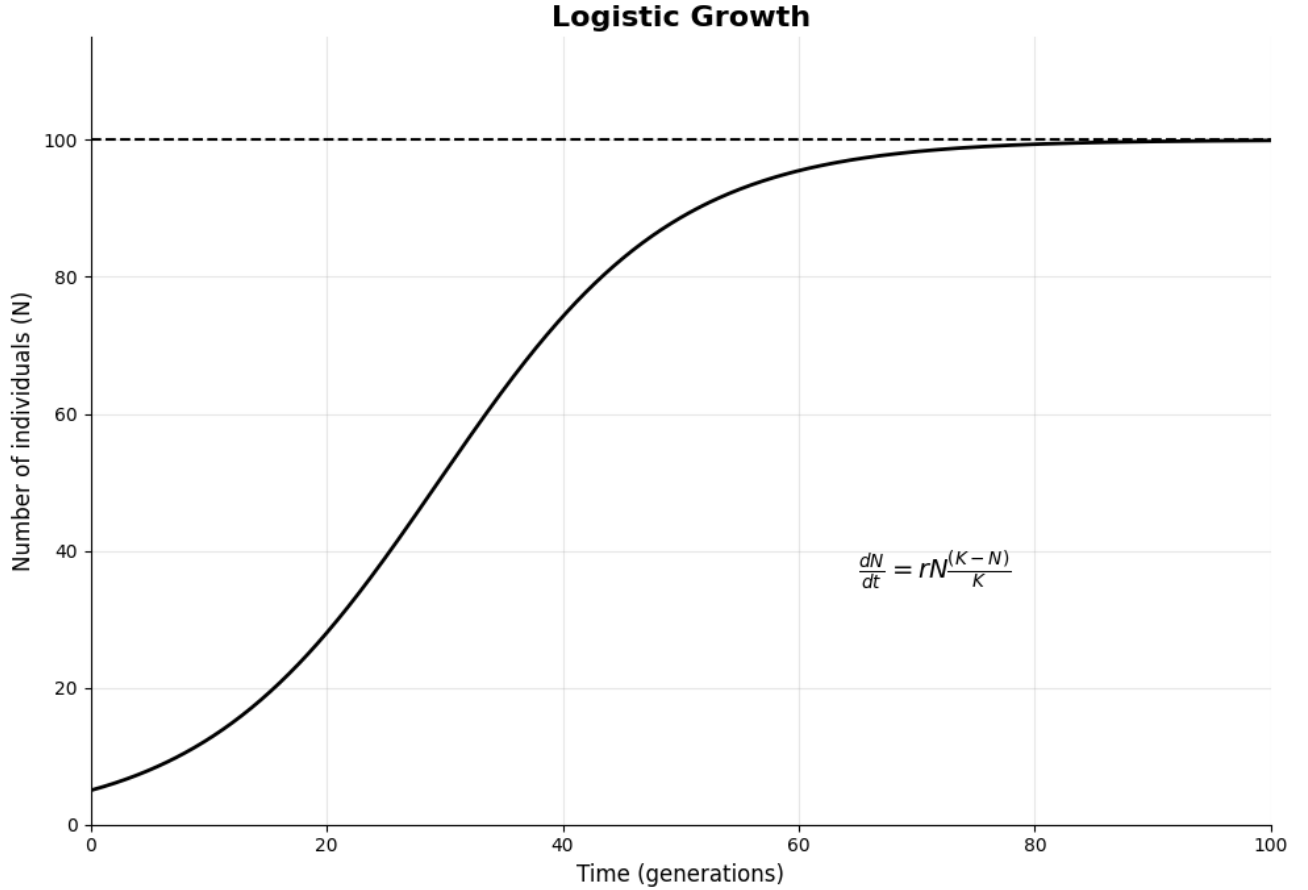


Figure 2: Logistic growth curve showing how population size N changes over time t with intrinsic growth rate r and carrying capacity K .

The model predicts rapid growth at low population levels, with growth slowing as N approaches K . Although real populations may fluctuate, it provides a useful baseline. Stokes (2012)

2.2 Two Species competition Lotka-Volterra Model

The Lotka-Volterra competition model extends the logistic model to two competing species. The model is written as:

$$\begin{aligned} \frac{dN_1}{dt} &= r_1 N_1 \left(1 - \frac{N_1 + \alpha_{12} N_2}{K_1}\right), \\ \frac{dN_2}{dt} &= r_2 N_2 \left(1 - \frac{N_2 + \alpha_{21} N_1}{K_2}\right), \end{aligned} \quad (2)$$

Here, K_i is the carrying capacity of species i , while α_{ij} measures the competitive effect of species j on species i . Depending on the competition strength and initial populations, the long-run outcomes may be coexistence, or dominance by one species. Gownaris and Zallek (2022); Murray (1993)

2.3 Harvesting and Management

Harvesting removes biomass from a fish population and can substantially alter its long run behavior. A simple harvested logistic model is:

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right) - H \quad (3)$$

where H denotes the harvest rate.

A common reference point in fisheries modelling is the Maximum Sustainable Yield (MSY), the largest harvest that can be taken from a fish population without causing long-term collapse.

Contributors (2019b)

For the harvested logistic model:

$$H_{MSY} = \frac{Kr}{4} \quad (4)$$

where H_{MSY} is the Maximum Sustainable Yield.

In practice, harvesting at the MSY may be risky, as small changes in environmental conditions, model parameters, or human error may make this level unsustainable. So it is useful to consider lower harvest amounts, such as the Maximum Economic Yield (MEY), the yield with the largest possible difference between total revenue and total cost of fishing, in other words, the yield that allows for the most profit long term. MEY is typically achieved at 80-90% of the MSY. *MSY -Maximum sustainable yield* (2012)

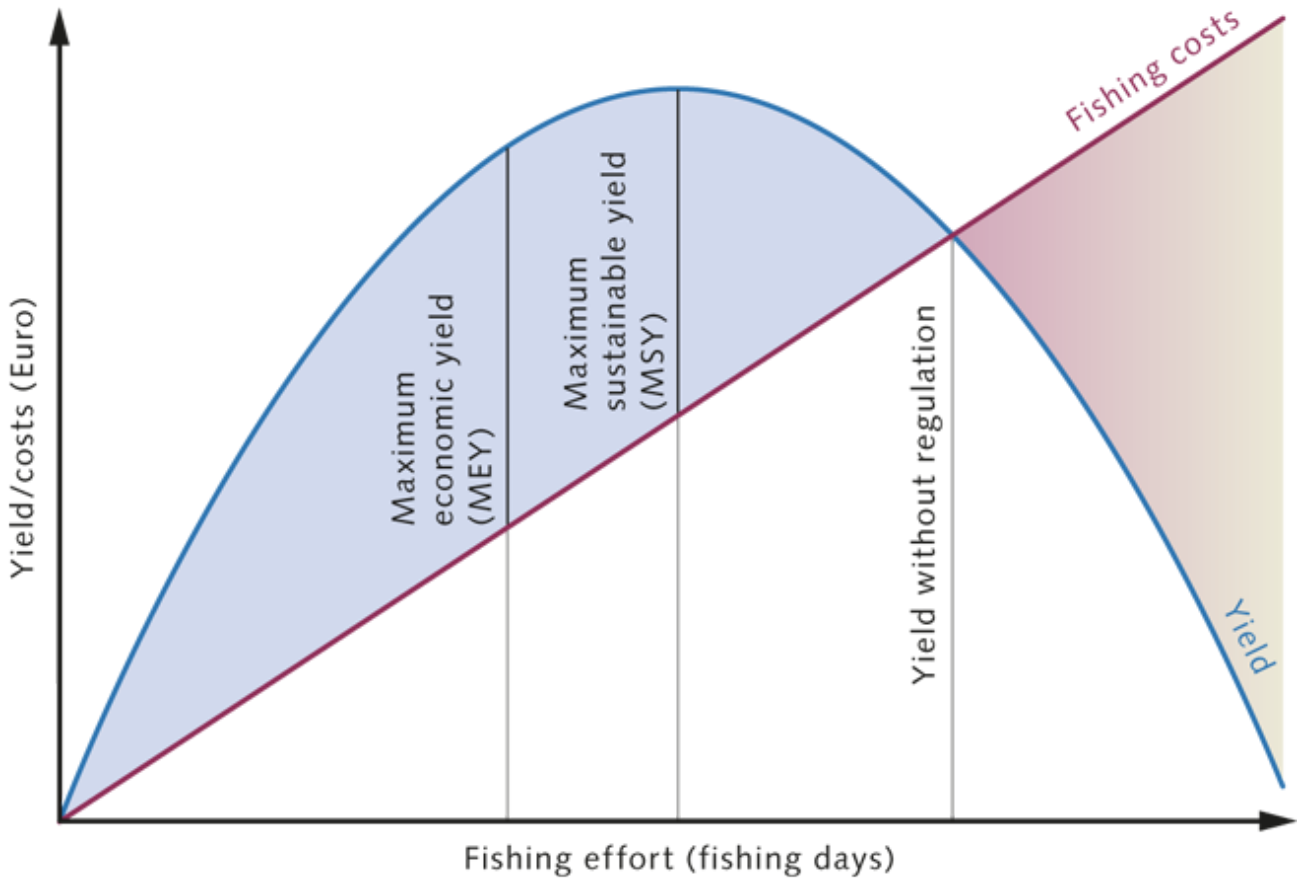


Figure 3: Relationship between fishing effort, yield, and profit showing the Maximum Sustainable Yield (MSY) and Maximum Economic Yield (MEY). Highlights why harvesting below the MSY can be more profitable long term.

The profits of fishing - World Ocean Review (2024)

2.4 Diffusion as Migration

Fish movement can be represented by diffusion, which provides a simple first approximation for migration in a PDE model. A spatial population model then takes the form:

$$\frac{\partial N}{\partial t} = f(N) + D \frac{\partial^2 N}{\partial x^2} \quad (5)$$

where $f(N)$ represents local population dynamics such as growth or harvesting, while D measures the rate of spatial movement Wan (2018)

Although real migration depends on environmental and biological factors, diffusion is sufficient here as a first approximation. *DAMS, FISH AND FISHERIES- Opportunities, challenges and conflict resolution* (2024)

3 Model Description and Boundary Form

In this section, nm is used to refer to nautical miles rather than nanometres.

3.1 Model Assumptions

The following assumptions have been made:

- Fish density depends only on offshore distance and time. Other factors, such as distance from the coast and temperature, do not affect it.
- Each species would follow logistic population growth if the other were absent
- Each species competes for shared resources
- There are only two species competing for resources; other species are not considered
- The species do not have a predator-prey relationship
- Environmental parameters such as growth rate, carrying capacity, competition coefficients and diffusion rates are constant
- Migration is represented by diffusion
- Harvest rates are chosen relative to the single species Maximum Sustainable Yield (MSY) harvest rate, and differ inside and outside the EEZ
- The competitive effect of species i affects species j , where $i \neq j$, is constant, and the model doesn't account for any potential changes in this effect
- No fish can enter or leave the system through the coastline so no flux boundary conditions are imposed

These assumptions produce a tractable PDE model for harvesting, competition and migration.

3.2 Spatial Domain and Policy Setup

The model uses a one-dimensional offshore coordinate s , measured from the coastline. We take $s = 0$ at the shore, $s_E = 200$ at the EEZ boundary, and $s_L = 600$ at the outer computational boundary. The domain is therefore divided into a more regulated region $0 \leq s \leq s_E$ and a less regulated region $s_E < s \leq s_L$. The 200-mile line is treated as an internal policy interface: harvesting changes across it, while diffusion remains continuous across the whole domain.

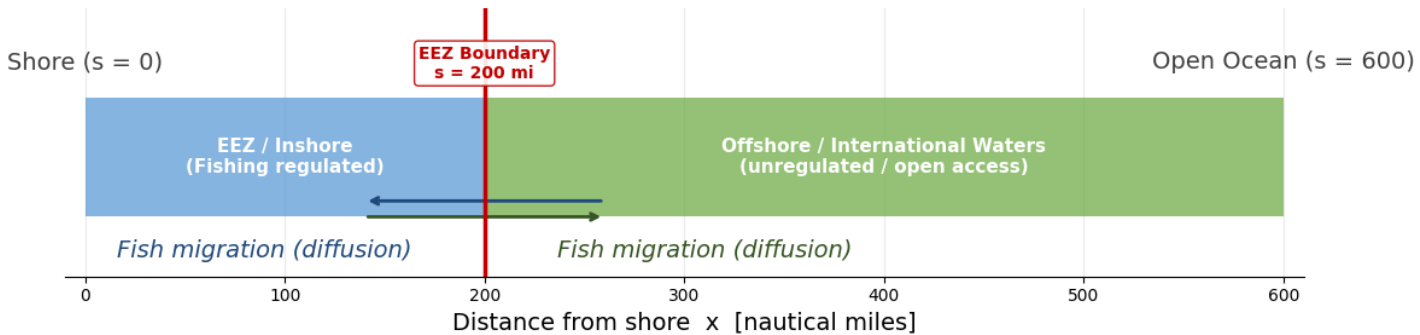


Figure 4: One-Dimensional offshore domain used in the PDE model. The coastline is at $s = 0$, the EEZ boundary is at $s = s_E = 200$ nm, and the outer computational boundary is at $s = s_L = 600$ nm. The harvest rate changes across the EEZ boundary, while diffusion remains continuous across the whole domain.

Figure 4, summarises the spatial setup. We can see that harvesting pressure is lower inside the EEZ, and higher outside it. The 200-mile limit serves as a policy interface for the harvest function, determining how much is harvested based off fish position

3.3 Harvest function and management zones

Harvesting is imposed as proportional population removal. For a single non-spatial population, this gives the logistic harvesting model

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right) - hN, \quad (6)$$

where h is the harvest rate

For $h < r$, the positive equilibrium satisfies

$$N^*(h) = K \left(1 - \frac{h}{r}\right). \quad (7)$$

So the sustainable yield, is therefore

$$Y(h) = hN^*(h) = hK \left(1 - \frac{h}{r}\right). \quad (8)$$

This is maximised at

$$h_{\text{MSY}} = \frac{r}{2}, \quad (9)$$

with the corresponding yield Y_{MSY}

$$Y_{\text{MSY}} = \frac{rK}{4}. \quad (10)$$

In the two-species model, these single-species MSY rates are used to set spatially varying harvest rates. For species $i \in \{1, 2\}$, we define

$$h_{i,\text{MSY}} = \frac{r_i}{2}. \quad (11)$$

The harvest rate for species i is

$$h_i(s) = \begin{cases} h_{i,\text{in}}, & 0 \leq s \leq s_E, \\ h_{i,\text{out}}, & s_E < s \leq s_L, \end{cases} \quad (12)$$

with

$$h_{i,\text{in}} = \eta_{i,\text{in}} h_{i,\text{MSY}}, \quad h_{i,\text{out}} = \eta_{i,\text{out}} h_{i,\text{MSY}}, \quad (13)$$

with multipliers

$$\eta_{i,\text{in}} \in [0.8, 0.9], \quad \eta_{i,\text{out}} \in [0.9, 1.0]. \quad (14)$$

Thus lower harvest inside the EEZ and higher outside of it.

The function $h_i(s)$ is discontinuous at $s = s_E$, but the 200-mile line is not treated as a PDE boundary; it acts only as an internal policy interface so diffusion remains continuous across the line.

3.4 Dimensional Single Species Benchmark Model

Before introducing competition, we use the single-species benchmark

$$\frac{\partial u}{\partial t} = ru \left(1 - \frac{u}{K}\right) - h(s)u + D \frac{\partial^2 u}{\partial s^2}, \quad (15)$$

where u is population density, $h(s)$ is the spatially varying harvest rate, and D is the diffusion coefficient. This model isolates the effect of spatially varying harvest under the EEZ policy.

3.5 Dimensional Competing-Species Model

The competing species model is:

$$\begin{aligned} \frac{\partial u_1}{\partial t} &= r_1 u_1 \left(1 - \frac{u_1 + \alpha_{12} u_2}{K_1}\right) - h_1(s)u_1 + D_1 \frac{\partial^2 u_1}{\partial s^2}, \\ \frac{\partial u_2}{\partial t} &= r_2 u_2 \left(1 - \frac{u_2 + \alpha_{21} u_1}{K_2}\right) - h_2(s)u_2 + D_2 \frac{\partial^2 u_2}{\partial s^2} \end{aligned} \quad (16)$$

where u_1 and u_2 are the species densities, α_{ij} are competition coefficients, and $h_i(s)$ are the spatially varying harvest rates. This extends the benchmark model by incorporating competition.

3.6 Non-Dimensional Single Species Benchmark Model

To reduce the number of parameters, we non dimensionalise the single species model as:

$$\frac{\partial x}{\partial \tau} = x(1 - x) - \lambda(\xi)x + \delta \frac{\partial^2 x}{\partial \xi^2} \quad (17)$$

with our scaling:

$$\begin{aligned} x &= \frac{u}{K}, & \xi &= \frac{s}{L}, & \tau &= rt, \\ L = s_L &= 600\text{nm}, & \lambda(\xi) &= \frac{h(L\xi)}{r}, & \delta &= \frac{D}{rL^2} \end{aligned} \quad (18)$$

where x is our dimensionless density, ξ is our dimensionless offshore distance, τ is our dimensionless time, $\lambda(\xi)$ is our dimensionless harvest rate function, δ is our dimensionless diffusion coefficient.

The offshore domain becomes

$$0 < \xi < 1, \quad \xi_L = 1, \quad \xi_E = \frac{s_E}{s_L} = \frac{1}{3}$$

3.7 Non-Dimensional Competing-Species Model

Similarly, the nondimensional competing-species model is:

$$\begin{aligned} \frac{\partial x}{\partial \tau} &= x(1 - x - \beta_{12}y) - \lambda_1(\xi)x + \delta_1 \frac{\partial^2 x}{\partial \xi^2}, \\ \frac{\partial y}{\partial \tau} &= \rho y(1 - y - \beta_{21}x) - \lambda_2(\xi)y + \delta_2 \frac{\partial^2 y}{\partial \xi^2}. \end{aligned} \quad (19)$$

with our scaling:

$$\begin{aligned} x &= \frac{u_1}{K_1}, & y &= \frac{u_2}{K_2}, & \xi &= \frac{s}{L}, & \tau &= r_1 t, \\ L = s_L &= 600\text{nm}, & \rho &= \frac{r_2}{r_1}, & \beta_{12} &= \frac{\alpha_{12}K_2}{K_1}, & \beta_{21} &= \frac{\alpha_{21}K_1}{K_2}, \\ \lambda_1(\xi) &= \frac{h_1(L\xi)}{r_1}, & \lambda_2(\xi) &= \frac{h_2(L\xi)}{r_1}, & \delta_1 &= \frac{D_1}{r_1 L^2}, & \delta_2 &= \frac{D_2}{r_1 L^2}. \end{aligned} \quad (20)$$

where x and y are our dimensionless densities for species one and two, respectively, ξ is our dimensionless offshore distance, τ is our dimensionless time, ρ is the ratio of intrinsic growth rates between the populations, β_{12} and β_{21} are our dimensionless competition coefficients, τ is our dimensionless time, $\lambda_1(\xi)$ and $\lambda_2(\xi)$ are our dimensionless harvest rate functions, and δ_1 and δ_2 are our dimensionless diffusion coefficients.

The nondimensional domain is again $0 < \xi < 1$ with $\xi_E = 1/3$.

3.8 Initial and Boundary Condition

Initial and boundary conditions are stated for the non-dimensional system.

For our single species:

$$x(\xi, 0) = x_0(\xi), \quad 0 < \xi < 1 \quad (21)$$

For our competing species model:

$$x(\xi, 0) = x_0(\xi), \quad y(\xi, 0) = y_0(\xi), \quad 0 < \xi < 1. \quad (22)$$

In our numerical experiments, $x_0(\xi)$ and $y_0(\xi)$ are smooth positive Gaussian-type profiles.

No-flux boundary conditions are imposed at both ends of the domain. For the single species:

$$\frac{\partial x}{\partial \xi}(0, \tau) = 0, \quad \frac{\partial x}{\partial \xi}(1, \tau) = 0, \quad \tau > 0. \quad (23)$$

For the competing species model:

$$\frac{\partial x}{\partial \xi}(0, \tau) = 0, \quad \frac{\partial x}{\partial \xi}(1, \tau) = 0, \quad \frac{\partial y}{\partial \xi}(0, \tau) = 0, \quad \frac{\partial y}{\partial \xi}(1, \tau) = 0, \quad \tau > 0. \quad (24)$$

These conditions ensure that diffusion redistributes biomass without gain or loss across the computational boundary. No additional boundary condition is imposed at $\xi = \xi_E$; the EEZ line acts only through the harvest function.

4 Numerical Method and Validation

4.1 Numerical Implementation

To numerically solve the model, we discretise the one-dimensional offshore domain using a uniform finite-difference grid. We will present the numerical scheme for the non-dimensional single-species benchmark model

$$\frac{\partial x}{\partial \tau} = x(1 - x) - \lambda(\xi)x + \delta \frac{\partial^2 x}{\partial \xi^2}, \quad 0 < \xi < 1, \quad \tau > 0, \quad (25)$$

with no-flux boundary conditions at $\xi = 0$ and $\xi = \xi_L = 1$. The competing-species system is treated in the same way, with the scheme applied to both species.

Since a computer can not store a whole continuous function exactly, we discretise the offshore domain using grid points

$$\xi_j = j\Delta\xi, \quad j = 0, 1, \dots, N-1, \quad (26)$$

where $\Delta\xi = 1/(N-1)$. Time is discrete as

$$\tau^n = n\Delta\tau, \quad n = 0, 1, 2, \dots, \quad (27)$$

and x_j^n denotes the numerical approximation to $x(\xi_j, \tau^n)$, solution at grid point j and time step n .

For interior grid points, the second derivative $\frac{\partial^2 x}{\partial \xi^2}$ is approximated using the standard second-order central difference

$$\left. \frac{\partial^2 x}{\partial \xi^2} \right|_j \approx \frac{x_{j+1}^n - 2x_j^n + x_{j-1}^n}{\Delta\xi^2}, \quad j = 1, \dots, N-2. \quad (28)$$

This measures local curvature so that if x_j^n is larger than its neighbours in the grid, diffusion will push it down, and if it is smaller, diffusion will pull it up. This is how we numerically represent spatial spreading.

Once we have approximated the spatial derivative, time integration is carried out using the explicit forward Euler method to update the solution forward in time, giving

$$x_j^{n+1} = x_j^n + \Delta\tau \left[x_j^n(1 - x_j^n) - \lambda(\xi_j)x_j^n + \delta \frac{x_{j+1}^n - 2x_j^n + x_{j-1}^n}{\Delta\xi^2} \right]. \quad (29)$$

So at each step we start from the current value x_j^n , compute growth, harvesting and diffusion, add a time step to get a new value x_j^{n+1}

The no-flux boundary conditions are implemented using Neumann boundary approximations. This gives

$$\left. \frac{\partial^2 x}{\partial \xi^2} \right|_0 \approx \frac{2(x_1^n - x_0^n)}{\Delta\xi^2}, \quad \left. \frac{\partial^2 x}{\partial \xi^2} \right|_{N-1} \approx \frac{2(x_{N-2}^n - x_{N-1}^n)}{\Delta\xi^2}. \quad (30)$$

These approximations enforce zero diffusive flux at both the coastline and the outer offshore boundary, so no biomass leaves the computational domain.

The same approach is used for the competing-species system, with the same scheme applied to each species separately. The time step is small enough to satisfy both a diffusion stability restriction and a reaction-harvest restriction, ensuring small errors, such as round-off or truncation, do not cause solutions to diverge in our simulations.

4.2 Validation

The numerical method is checked against benchmark behaviour before being used for the main simulations. In particular, we verified that the no-flux boundary conditions are implemented correctly and that the numerical solution is consistent with the underlying reaction-diffusion model.

In the single-species case, the numerical solution is compared with the known logistic behaviour to check that the reaction and harvesting terms have been implemented correctly. In addition, the diffusion term is checked to confirm that it redistributes biomass within the domain without biomass entering or leaving through the boundaries.

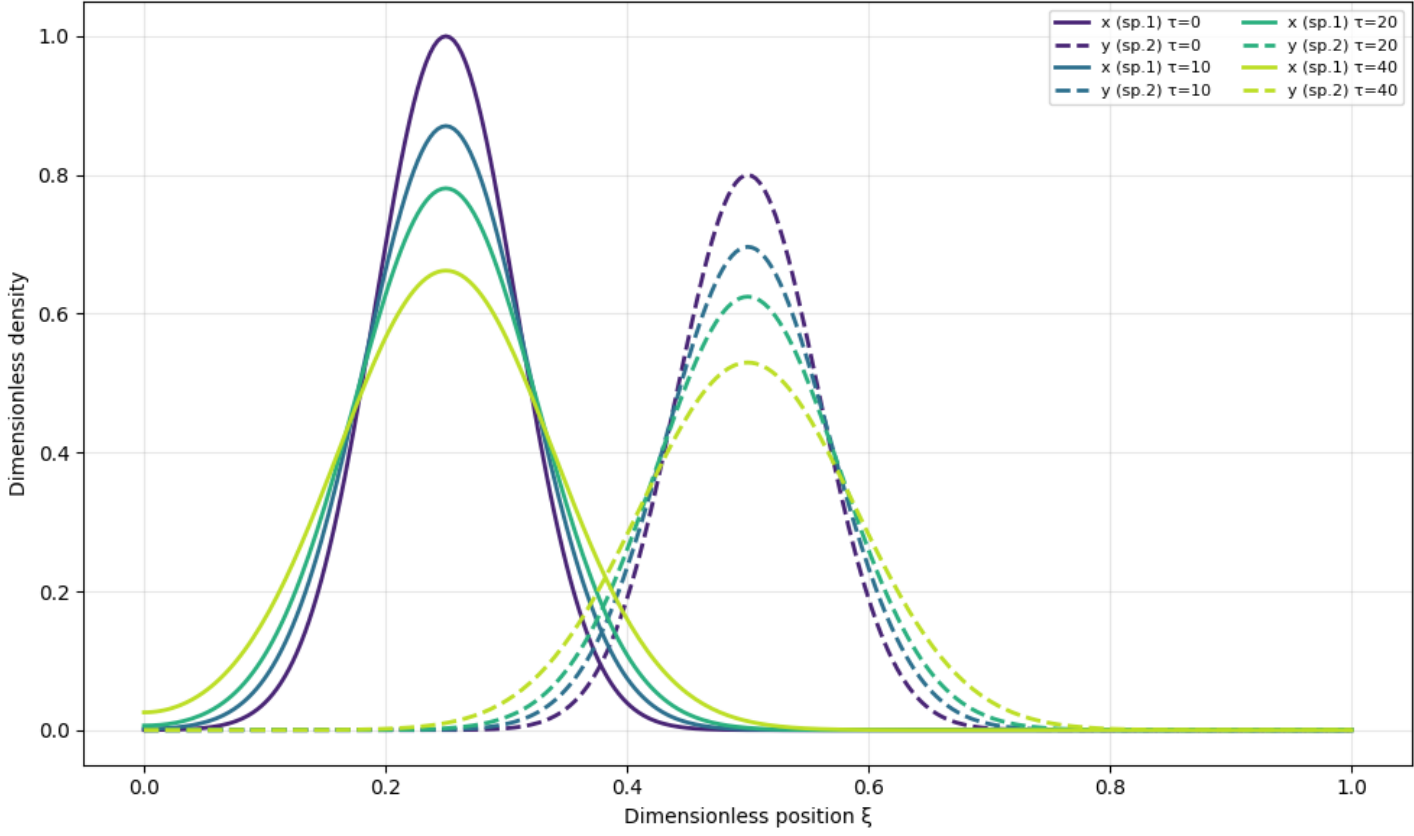


Figure 5: Validation of the diffusion term using a Gaussian initial profile under no flux boundary conditions. The density spreads and flattens over time while total biomass is conserved, which is consistent with diffusion under no-flux boundary conditions.

In Figure 5, we can see that diffusion acts as we would expect, spreading out a Gaussian initial profile, with the total mass conserved over time, confirming that the diffusion is correctly implemented. The solution flattens out as time progresses, which is consistent with diffusion acting to reduce spatial gradients in biomass density.

These checks confirm that the numerical method captures the intended behaviour before carrying out a deeper analysis of our model.

5 Baseline Results

Unless otherwise stated, the baseline simulations use the policy setup introduced in Section 3.2. The 200-mile line is at $\xi = \frac{1}{3}$. The single-species benchmark is tested with smooth Gaussian initial profiles centred at various locations in our schematic, and the same is done for the two-species form. Simulations use a stochastic harvest schedule generated at the start of the program and then applied over the simulation. We start our yearly harvest at $\tau = 10$,

5.1 Single Species Results

Before introducing competition, we use the single-species model from Section 3.4 to isolate the effect of the two-zone EEZ policy. The diffusion parameter is fixed throughout the baseline runs and, with the dimensional values $r = 0.5$, $D = 10$, and $L = 600$, gives dimensionless diffusion

$$\delta = \frac{D}{rL^2} \approx 5.56 \times 10^{-5}.$$

Our h_{MSY} is

$$\lambda_{MSY} = \frac{h_{MSY}}{r} = \frac{1}{2}$$

5.1.1 Baseline Spatiotemporal Evolution

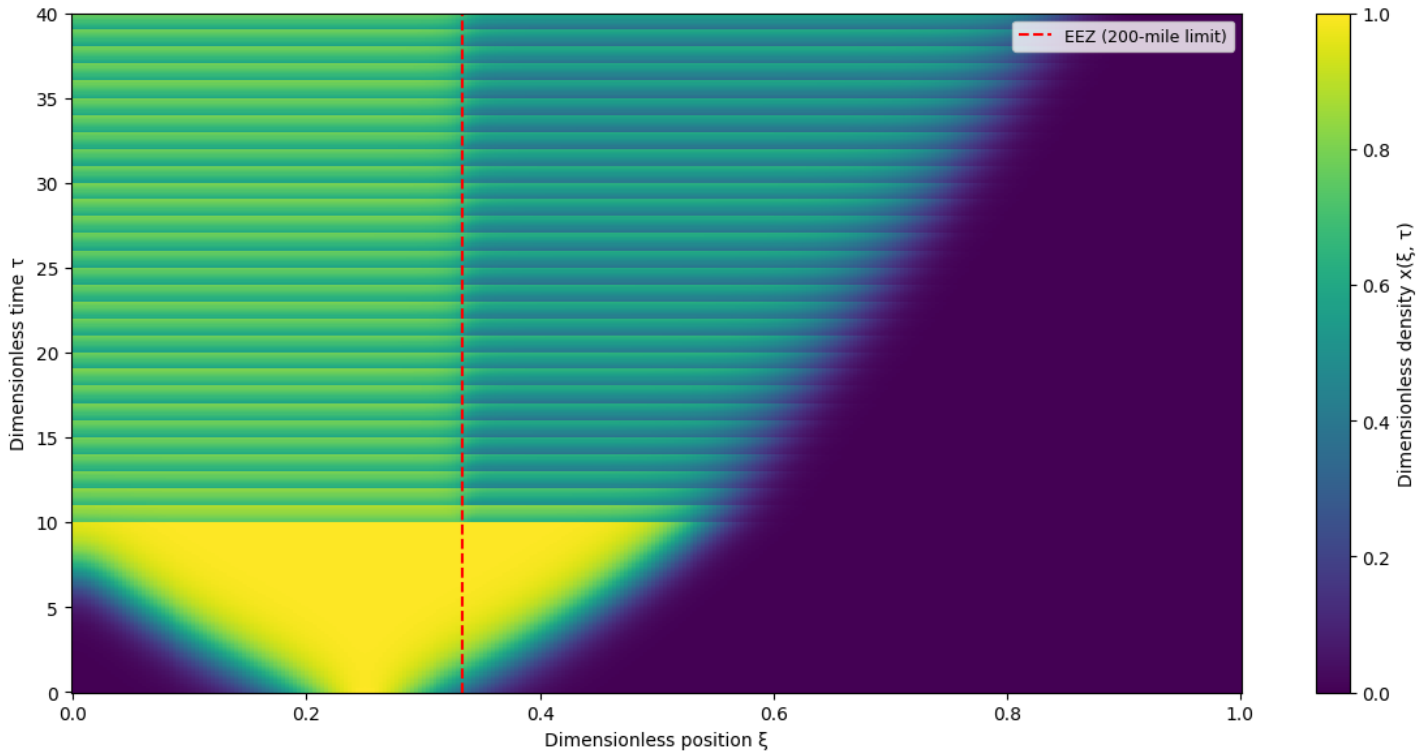


Figure 6: Spatiotemporal evolution of the single species model under the two-zone 200-mile fishing policy. The initial condition is a smooth Gaussian pulse, the dimensionless diffusion parameter is $\delta \approx 5.56 \times 10^{-5}$. The dashed vertical line indicates the EEZ boundary at $\xi = \frac{1}{3}$.

In Figure 6, we can see that early on, the population is concentrated near the initial Gaussian pulse, but over time, the biomass migrates across the domain, making the harvest difference more pronounced. Inside the EEZ, we observe lower harvest rates, resulting in higher fish density than outside the EEZ. We can also see that overall biomass is increasing over time as the population grows and spreads through diffusion.

5.1.2 Selected Timeshot Snapshots

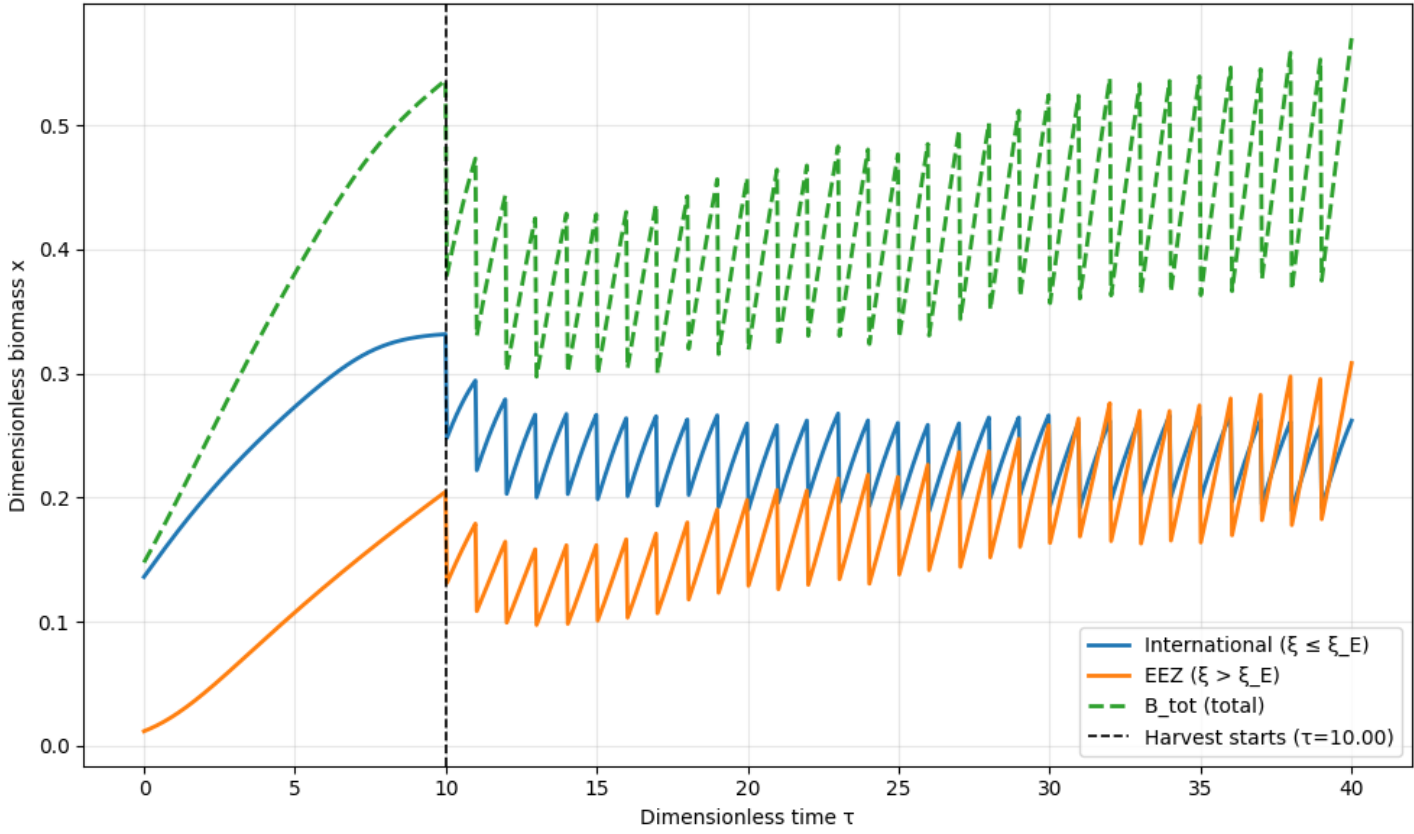


Figure 7: Temporal evolution of the biomass inside the EEZ, outside the EEZ and total under the two-zone 200-mile fishing policy. Dash vertical line indicates when harvesting starts at $\tau = 10$

In Figure 7, we can see that before harvest, the biomass increases both inside and outside the EEZ due to diffusion and the logistic growth. Once harvest starts at $\tau = 10$, we see a significant drop in biomass all around the domain, with the fish inside reaching the shore without diffusing any closer, whereas since the fish outside have more areas to diffuse to, they eventually have a higher population despite being exposed to higher harvest rates. If the limit were raised to $\xi = 300$, we would expect the fish inside to have a higher long-term population due to having an equal maximum size to diffuse to, like the fish outside, as well as being harvested less.

5.1.3 Robustness to Start Position

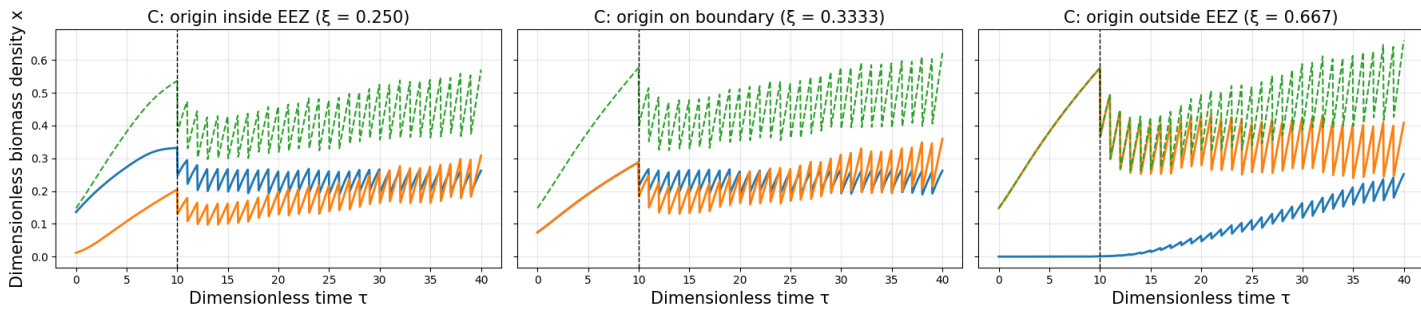


Figure 8: Temporal evolution of the biomass under the two-zone 200-mile fishing policy with different initial Gaussian pulse positions. The dashed vertical line indicates when harvesting starts at $\tau = 10$.

The green line is the total mass, the blue line is mass inside the EEZ, and the orange line is mass outside the EEZ.

In Figure 8, we can see that the initial position has a significant effect on the biomass. We can see that the region with the initial Gaussian bump has a higher population and reaches equilibrium much more quickly. In the long run, we see that the solutions converge to a similar biomass. We can see that when the Gaussian Bump is outside the EEZ, it takes longer for the biomass to cross into the other zone in comparison to when the Gaussian bump is in the EEZ

5.2 Competing Species Results

We now turn to the competing-species model from Section 3.5 to examine how competition alters EEZ dynamics. Unless otherwise stated, the baseline runs use these competition coefficients,

$$\beta_{12} = 0.4, \quad \beta_{21} = 0.5,$$

which places the system in the coexistence regime. The initial Gaussian profiles are centred at $\xi = 1/6$ for species 1 and $\xi = 1/4$ for species 2.

5.2.1 Baseline Spatiotemporal Evolution

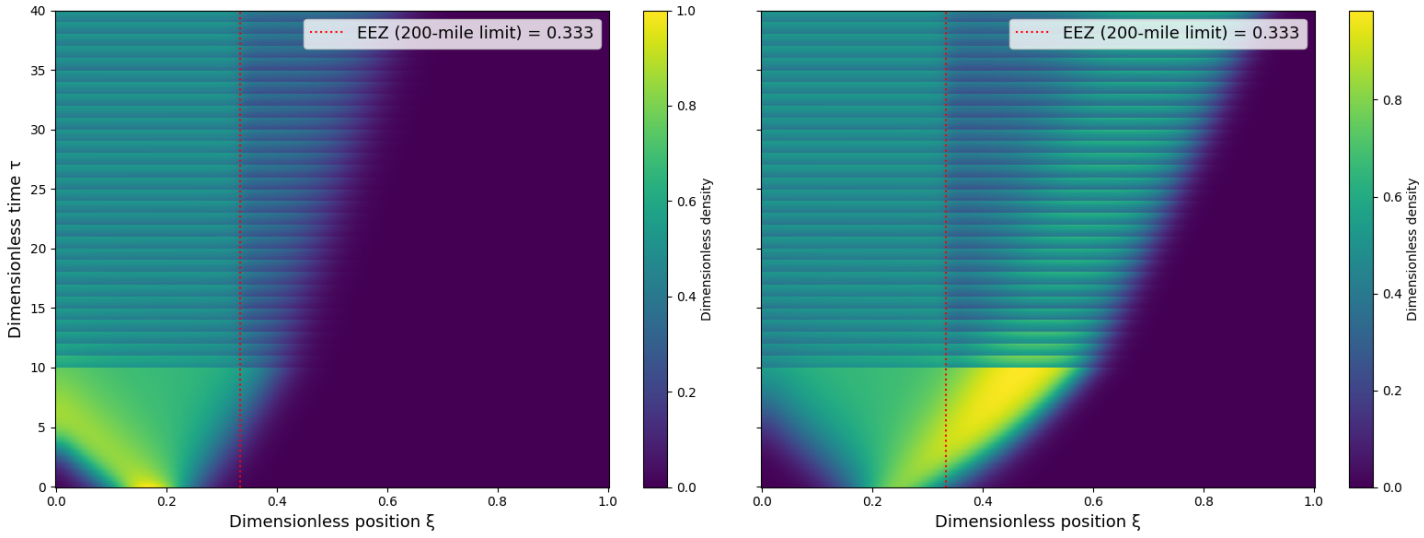


Figure 9: Spatiotemporal evolution of the competing-species model under the baseline two-zone 200-mile fishing policy. The dashed vertical line indicates the EEZ boundary at $\xi = \frac{1}{3}$.

In Figure 9, we can see that early on, the populations are concentrated close to each other, but then diffuse more widely in different directions, peaking in different directions because they are less affected by competition the further they are. Even though species 2 is experiencing more harvest, its peak density is farther from species 1 because it experiences less competition from that population. As species 2 starts closer to the limit, a larger proportion of its population spreads outside the EEZ than for species 1.

5.2.2 Selected Timeshot Snapshots

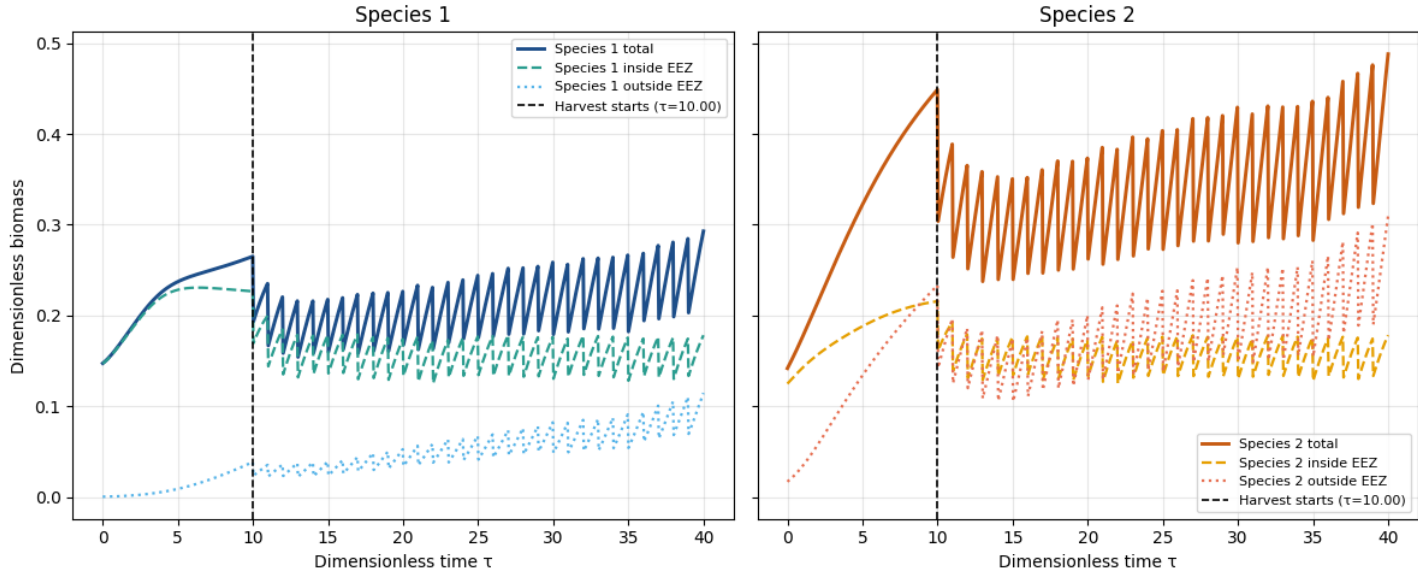


Figure 10: Temporal evolution of the species 1 and 2 inside the EEZ, outside the EEZ and total under the two-zone 200-mile fishing policy. Dash vertical line indicates when harvesting starts at $\tau = 10$

In Figure 10, we can see that before harvest, both populations grow and spread out, with species 2 having higher biomass because it starts closer to the limit, allowing it to grow farther from the shore. Once harvest starts at $\tau = 10$, we see a significant drop in biomass all around. Similar to the single species case, although the second species tends to experience a higher harvest, they still have a higher population due to having more room to grow outside the EEZ this is while the peak density is in the opposite direction from species 1 as seen from the previous heatmap.

5.2.3 Robustness to Start Position

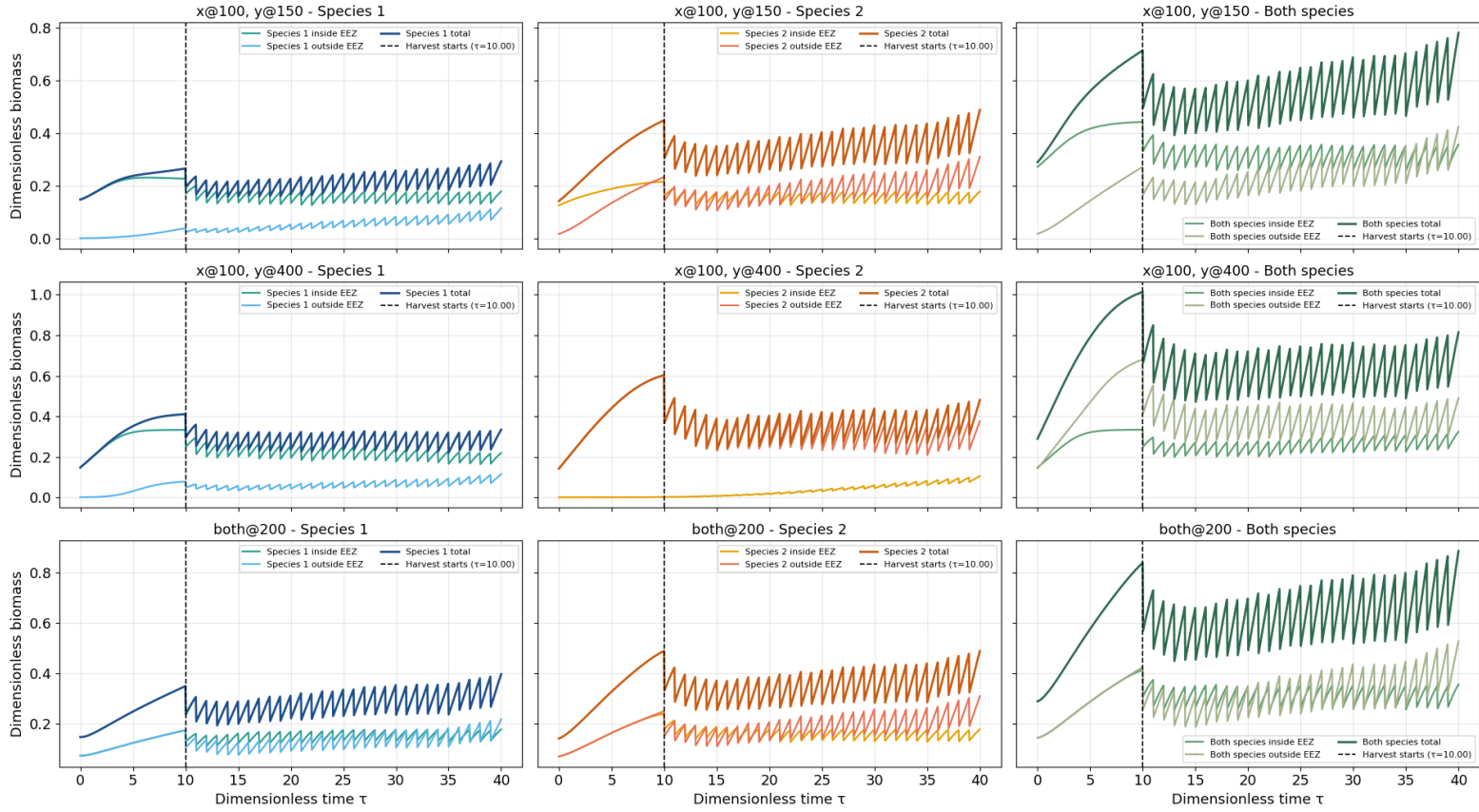


Figure 11: Temporal evolution of the biomass under the two-zone 200-mile fishing policy with varying initial Gaussian pulse positions to highlight how different start positions affect the long term biomass. The dashed vertical line indicates when harvesting starts at $\tau = 10$.

In Figure 11, we can see that the main effect of changing the initial arrangement is not whether the equilibrium type changes from coexistence, but rather how the biomass is split between inside and outside the EEZ. To analyse this further, we will introduce a graph describing the biomass percentage at varying start positions.

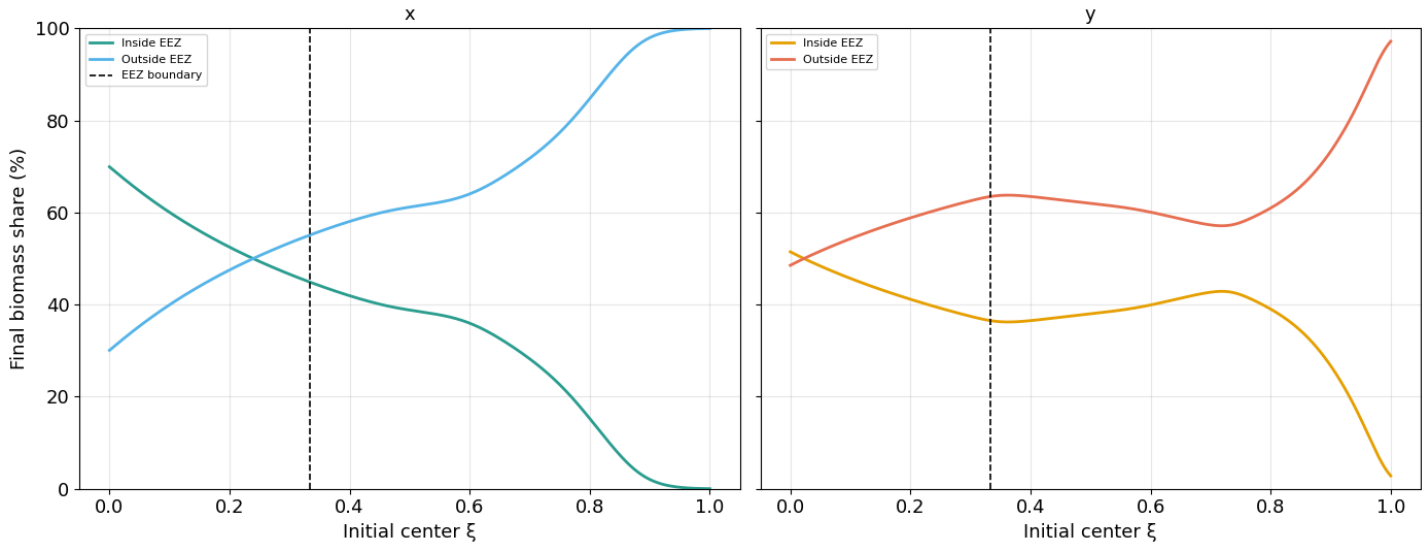


Figure 12: Percentage of biomass inside the EEZ at $\tau = 100$ under the two-zone 200-mile fishing policy with different initial Gaussian pulse positions highlighting how initial positions affect the distribution. The dashed vertical line indicates the EEZ boundary at $\xi = \frac{1}{3}$.

In Figure 12, we can see that despite competition the closer the Gaussian bump to the coastline gets to the coastline, the higher percentage of a species will reside inside the EEZ, whereas the further you get, the higher the percentage of the population resides outside the EEZ. Additionally, we can see that when populations are equal inside and outside, this happens when the Gaussian bump is inside the EEZ.

6 Sensitivity Analysis

6.1 Sensitivity Analysis Framework

To assess sensitivity, one parameter is varied at a time while all others are held fixed. For each parameter p , the non-dimensional PDE system is first solved for the baseline case up to $\tau_{\text{end}} = 50$, and the resulting long-time output measures are recorded. Here we use the equilibrium biomass:

$$x^*, y^*$$

Each parameter is varied by $\pm 10\%$, giving values $0.9p$ and $1.1p$. We had also established baseline values to support our non-dimensionalisation. We will estimate the sensitivity coefficients using the central finite difference formulas:

$$S_x(p) = \frac{x^*(1.1p) - x^*(0.9p)}{0.2}, \quad S_y(p) = \frac{y^*(1.1p) - y^*(0.9p)}{0.2} \quad (31)$$

These coefficients give us an estimate of how strongly long-term biomass responds to changes in each parameter, with larger absolute values indicating greater change.

6.2 Individual Parameter Sensitivity

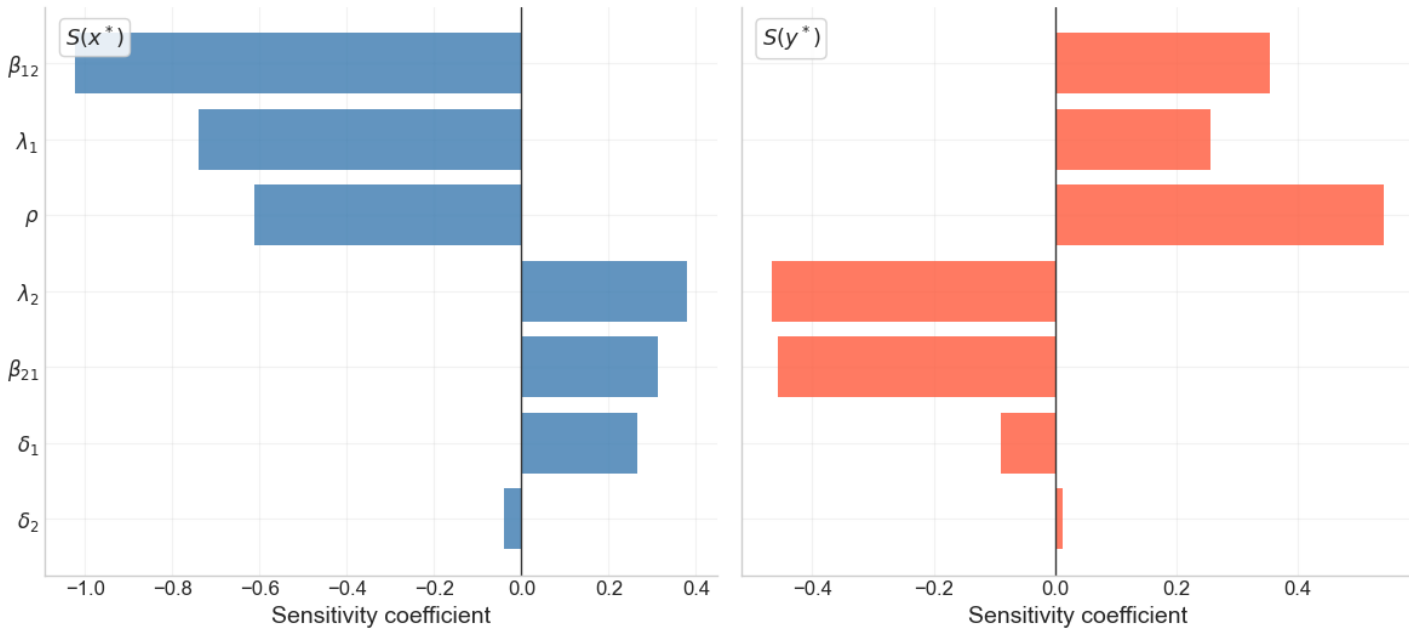


Figure 13: Tornado plot of one-at-a-time sensitivity coefficients for the long term competing species biomasses x^* and y^* at $\tau = 100$ computed by $\pm 10\%$ perturbations about the baseline parameters.

From Figure 13, we can see that the biomass of both populations is most sensitive to changes in the competition coefficients β_{ij} and harvest rate λ_i . The competition coefficients have a significant impact as the stronger the competition is from an opposing species, the more the species will be suppressed. Similarly, the harvest rate has a significant impact as well as the higher it is the more suppressed a species gets, also if one species is more suppressed, the other species is gets to experience more growth due to experiencing less competition. There is a clear mirror effect between the species:

- An increase in β_{12} suppresses species 1 and benefits species 2, while an increase in β_{21} suppresses species 2 and benefits species 1.
- An increase in λ_1 suppresses species 1 and benefits species 2, while an increase in λ_2 suppresses species 2 and benefits species 1.

Additionally, we see that diffusion δ_i does not have a significant effect on the system's outcome. We can also see that both are significantly affected by a change in our growth rate ratio ρ , with species 2 benefiting more from a higher relative growth rate.

Since we can see a mirror effect between the species, we can focus the rest of our analysis on

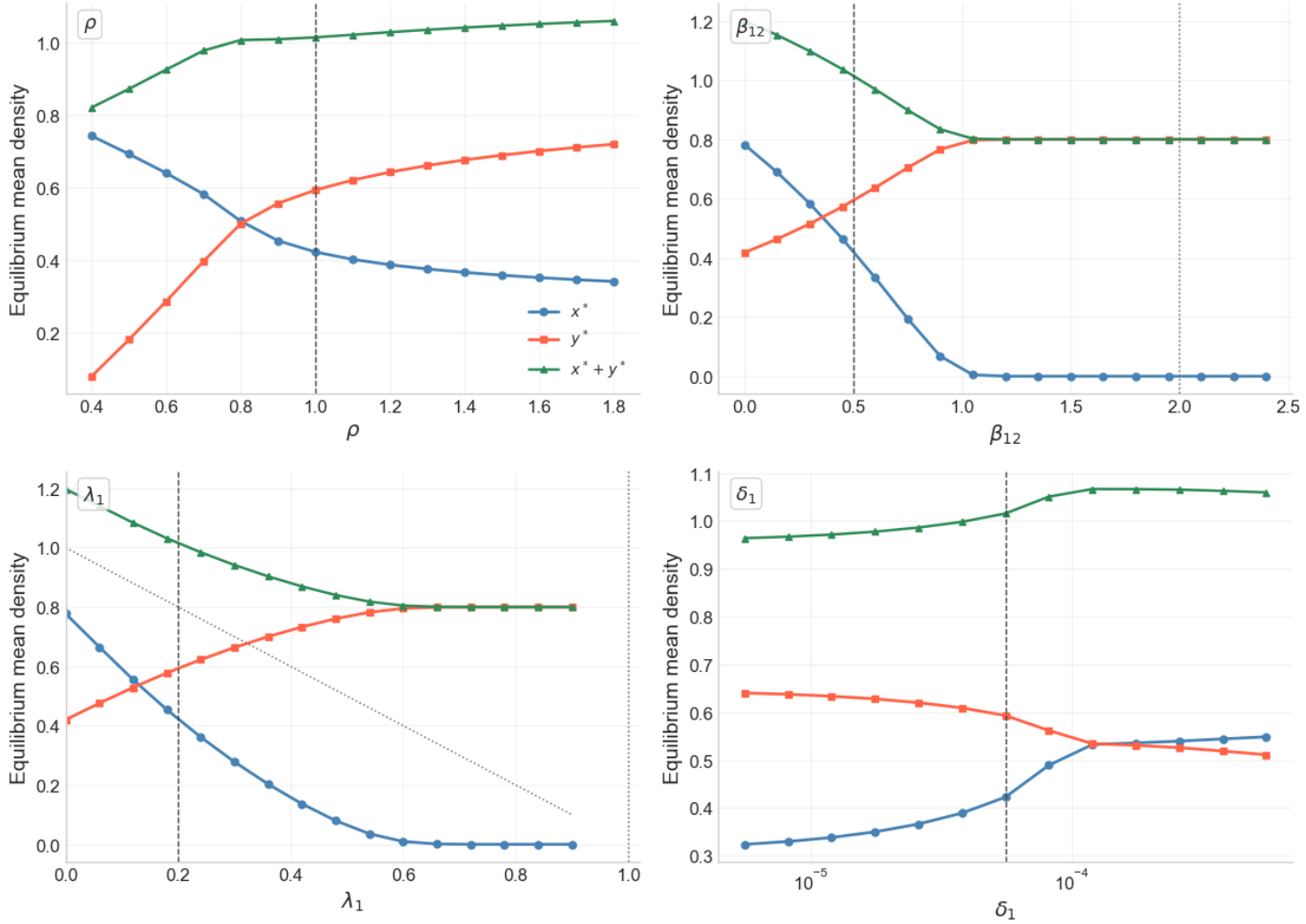


Figure 14: Equilibrium biomasses of species 1 and species 2, responses to one at a time parameter variations. Each line shows density at $\tau = 100$. Faded vertical lines indicate the baseline parameter values.

By varying our parameters, we can see that the trends are consistently monotonic but differ in strength and shape. Increasing ρ produces a gradual shift, where x^* decreases and y^* increases as species 2 benefits from an increased ratio in growth rates. We can also see a slight increase in population size, indicating that the competitive advantage species 2 gains increases the total population. In contrast, increasing β_{12} produces a much sharper shift, where x^* decreases and y^* increases as species 1 experiences more competition from species 2. We can also see a significant decrease in total population, indicating that stronger competition from species 2 suppresses species 1, reducing the total population and changing the equilibrium type. Increasing λ_1 produces a similar sharp shift, where x^* decreases and y^* increases as species 1 experiences more harvest. We can also see a significant decrease in total population, indicating that the higher harvest of species 1 significantly suppresses it, which reduces the total population. The increase in species 2 indicates that they benefit from fewer species 1 being in the system. Increasing δ_1 has a less significant effect than the other parameters, where larger diffusion allows population 1 to grow more, thereby decreasing the amount of species 2 due to its greater likelihood of facing competitors as species 1 is more diffused across the system.

6.3 Outside the EEZ

Since outside the EEZ is much more unpredictable and unregulated by us, it is worth exploring what can occur when we use a harvest different from the baseline. We will still keep the inside fixed as those are the waters we

can manage.

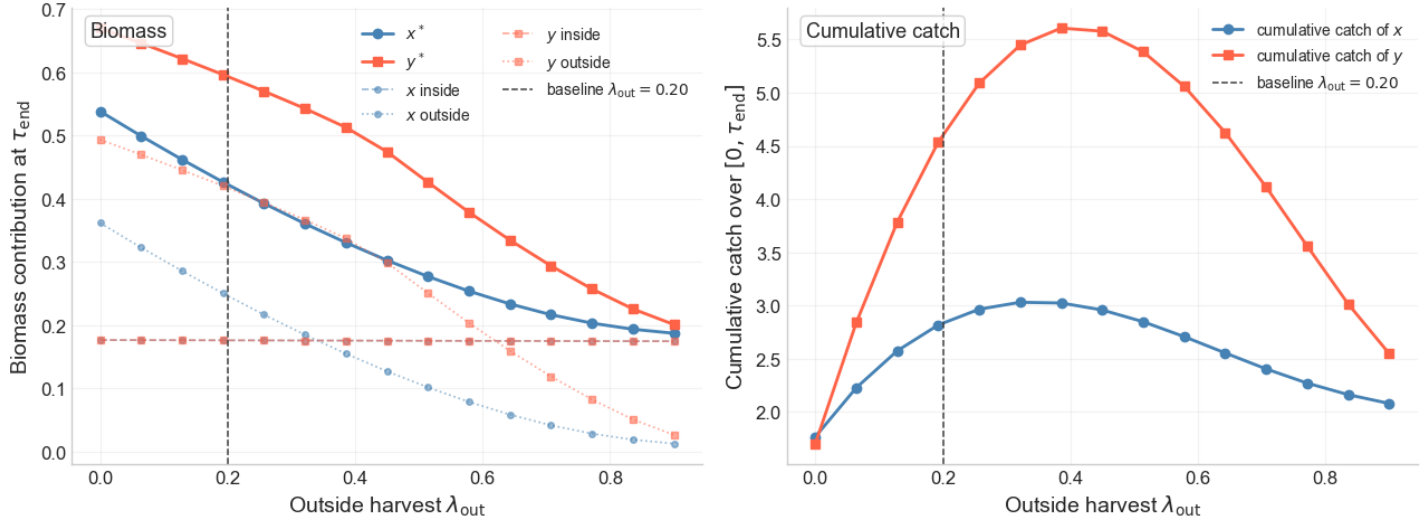


Figure 15: Effect of varying offshore harvest rate λ_{out} on the long-term biomass on long term biomass and cumulative catch. Faded vertical line indicate what was used in the baseline simulations. $\tau_{end} = 100$

As expected, we can see that increasing the harvest rate outside the EEZ leads to long-term decreases in biomass due to greater pressure on fish populations. On the right, we can see a hump shape: a higher harvest allows more fish to be caught overall, up to a certain point, $\lambda_{out} = 0.4$. After this, we can see that cumulative catch decreases, which is due to populations going instinctively beforehand. The figure highlights a clear tradeoff between exploitation and conservation, showing that simply maximising or minimising harvest does not necessarily lead to the best outcomes for fishermen in the long run.

7 Conclusion

This report used a one dimensional reaction-diffusion models to study how the 200-mile fishing limit affects fish population under spatially varying harvest. We had considered a single species benchmark model and a two species competition model, with lower harvest inside the EEZ and higher harvesting outside it.

Our findings that that the 200-mile limit can significantly affect both the spatial distribution of biomass as well as the long-run population outcomes. In the single species case, lower harvest supports greater biomass in the long run in contrast to outside of it, although migration allows biomass to move into more heavily harvested regions. In the competing species case, coexistence remains possible under the baseline parameters, but the balance between the two species depends strongly on competition strength, harvest rates, and initial spatial arrangement. The sensitivity analysis also showed that the model is most responses to changes in the competition coefficient and harvest intensity whilst diffusion played a weaker role for the baseline parameter set. Varying offshore harvest also revealed a clear trade-off between conservation and cumulative catch, where we can see that beyond a certain level, increasing harvest outside the EEZ reduces both biomass and long-term yield.

The model is a simplification of real world dynamics, as it is one dimensional, assumes constant parameters, migration is determined by a diffusion constant alone and only two species are included. In addition, the EEZ is treated as a smooth boundary, whereas in real life this may not be the case.

Future work can explore the effect of a more accurate representation of the system through the use of a 2D-model or extra boundaries. Additionally, we can consider the effect of different seasons or consider more fish species.

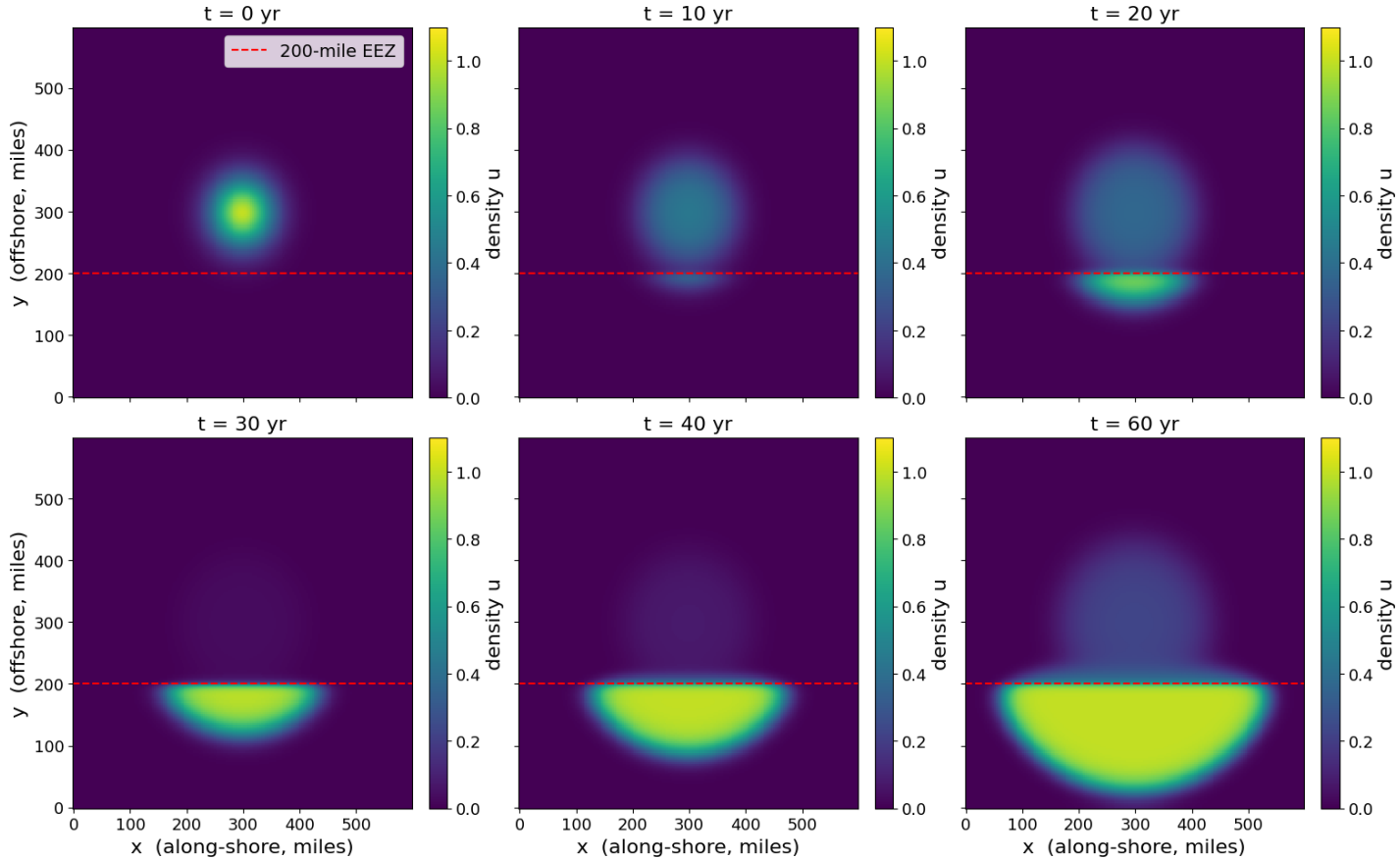


Figure 16: 2D Spatiotemporal evolution of the single species model under the two-zone 200-mile fishing policy shown through a heatmap. The dashed vertical line indicates the EEZ boundary at 200-nautical-miles.

In Figure 16, we can see how we can represent the spatiotemporal dynamics of the system through a 2D system, where we let our fish species start outside the EEZ, and as they are more prone to harvest they struggle to sustain a long term population. We can see the density going down but once fish start to migrate into the EEZ, they are able to sustain a long term population, due to the lower harvest.

A MSY Derivation

A.1 Single MSY Derivation

From Contributors (2019b) we can see that the MSY of a single species is achieved when the population size is half the carrying capacity, $N = \frac{K}{2}$. This is because if we take our logistic equation 1:

$$\begin{aligned}\frac{dN}{dt} &= rN \left(1 - \frac{N}{K}\right) - H \\ H(N) &= rN \left(1 - \frac{N}{K}\right) = rN - \frac{rN^2}{K}\end{aligned}$$

We can find the maxima of this with:

$$\begin{aligned}\frac{dH}{dN} &= r - \frac{2rN}{K} \\ N &= \frac{K}{2}\end{aligned}$$

We can then put this back into our H equation:

$$H\left(\frac{K}{2}\right) = \frac{rK}{4} \tag{32}$$

Thus, we have $\frac{rK}{4}$ as our MSY.

B Simulation Results

B.1 Tornado Plot Results

Table 1: One-at-a-time sensitivity results for the long-time biomass outputs under $\pm 10\%$ perturbations of the baseline parameter values.

Parameter	Baseline	$x^*(-10\%)$	$x^*(+10\%)$	$y^*(-10\%)$	$y^*(+10\%)$	$S(x^*)$	$S(y^*)$
β_{12}	0.50000	0.46479	0.37854	0.57248	0.61452	-1.02118	0.35436
λ_1	0.20000	0.45402	0.39161	0.57777	0.60811	-0.73888	0.25574
ρ	1.00000	0.45352	0.40183	0.55664	0.62078	-0.61204	0.54072
λ_2	0.20000	0.40648	0.43857	0.62074	0.56544	0.37982	-0.46613
β_{21}	0.50000	0.40951	0.43595	0.61938	0.56533	0.31308	-0.45562
δ_1	0.00006	0.41163	0.43407	0.59825	0.58761	0.26560	-0.08968
δ_2	0.00006	0.42440	0.42110	0.59222	0.59373	-0.03898	0.01260

The tables here aid in our sensitivity analysis in Section 6 by providing the raw data used to compute the sensitivity coefficients and the offshore harvest sweep.

B.2 Outside The EEZ Results

Table 2: Raw outputs from the offshore harvest-rate sweep used in Section 6.

λ_{out}	x_{tot}^*	y_{tot}^*	x_{in} biomass	x_{out} biomass	y_{in} biomass	y_{out} biomass	Cum. catch x	Cum. catch y
0.00000	0.53804	0.66963	0.17645	0.36159	0.17645	0.49318	1.76629	1.70130
0.06429	0.49910	0.64592	0.17631	0.32279	0.17631	0.46961	2.22734	2.83966
0.12857	0.46179	0.62140	0.17616	0.28563	0.17616	0.44524	2.57434	3.78520
0.19286	0.42618	0.59604	0.17602	0.25016	0.17602	0.42002	2.81688	4.53599
0.25714	0.39235	0.56978	0.17587	0.21648	0.17587	0.39391	2.96512	5.09041
0.32143	0.36039	0.54242	0.17572	0.18467	0.17572	0.36669	3.02998	5.44732
0.38571	0.33040	0.51259	0.17558	0.15483	0.17558	0.33701	3.02320	5.60704
0.45000	0.30252	0.47420	0.17543	0.12710	0.17543	0.29876	2.95755	5.57725
0.51429	0.27692	0.42591	0.17528	0.10163	0.17529	0.25062	2.84701	5.38460
0.57857	0.25381	0.37854	0.17514	0.07867	0.17515	0.20338	2.70689	5.05901
0.64286	0.23354	0.33442	0.17500	0.05854	0.17502	0.15940	2.55361	4.62543
0.70714	0.21653	0.29384	0.17486	0.04167	0.17488	0.11895	2.40348	4.11289
0.77143	0.20317	0.25725	0.17472	0.02845	0.17475	0.08250	2.27024	3.55890
0.83571	0.19354	0.22564	0.17459	0.01894	0.17462	0.05101	2.16184	3.01430
0.90000	0.18712	0.20096	0.17448	0.01264	0.17450	0.02646	2.07897	2.54800

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